Information Processing and Management 64 (2027) 105071

Contents lists available at ScienceDirect

Information Processing and Management

journal homepage: www.elsevier.com/locate/ipm

KG-ACE: Knowledge Graph Alignment and Consistency Enhancement for medical reasoning with large language models☆

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ARTICLE INFO

Keywords:
Large language models
Medical knowledge graphs
Medical reasoning
Alignment verification
Reasoning correction

ABSTRACT

Large language models (LLMs) can answer complex medical questions, but their intermediate reasoning may contain unsupported relations, omitted clinical constraints, or contradictions with established medical knowledge. We present Knowledge Graph Alignment and Consistency Enhancement (KG-ACE), a post-generation framework for verifying and revising medical reasoning produced by LLMs KG-ACE contains three components. First, a dual-layer medical knowledge graph separates standardized medical concepts in the Ontology Knowledge Graph from clinical rules in the Consistency Knowledge Graph. Second, a three-level verification mechanism evaluates concept validity, relation validity, and logical coherence. Third, a benefit-driven correction engine accepts a revision only when the recomputed alignment score improves by a specified margin. This design verifies the reasoning process without constraining the initial generation. Experiments on MedQA, MedMCQA, GenMedGPT-5k, and DDXPlus show that KG-ACE outperforms prompting-based, knowledge-graph-based, retrieval-based, and domain-pretrained baselines across all reported metrics. The results indicate that explicit process-level verification can complement knowledge augmentation by improving answer accuracy, reasoning stability, and the traceability of intermediate decisions.

1. Introduction

Large language models (LLMs) have expanded the range of natural language tasks that can be addressed with limited task-specific supervision (Brown et al., 2020; Ouyang et al., 2022). In medicine, models such as Med-PaLM have achieved strong performance on medical question-answering benchmarks, and LLMs are increasingly studied for clinical decision support, medical education, and patient communication (Singhal et al., 2023a; Thirunavukarasu et al., 2023). These capabilities are valuable in clinical problems that require information to be combined across symptoms, diseases, medications, and temporal events.

Strong benchmark performance does not, however, guarantee a sound reasoning process. Medical LLMs can produce fluent explanations that contain unsupported inferential steps, conflicting conclusions, or recommendations that violate clinical constraints. Existing work has primarily improved the knowledge available to the model through retrieval, data augmentation, or domain-specific

☆ This study was supported by the Suzhou Municipal Health Commission under Grant LCZX202304; the National Natural Science Foundation of China under Grants 62572335 and 62572336; the Natural Science Research of Jiangsu Higher Education Institutions of China under Grant 23KJA520011; and the Key Technologies Research and Development Program under Grant 2023YFF0725002.

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https://doi.org/10.1016/j.ipm.2026.105071
Received 9 April 2026; Received in revised form 20 July 2026; Accepted 20 July 2026
Available online 29 July 2026
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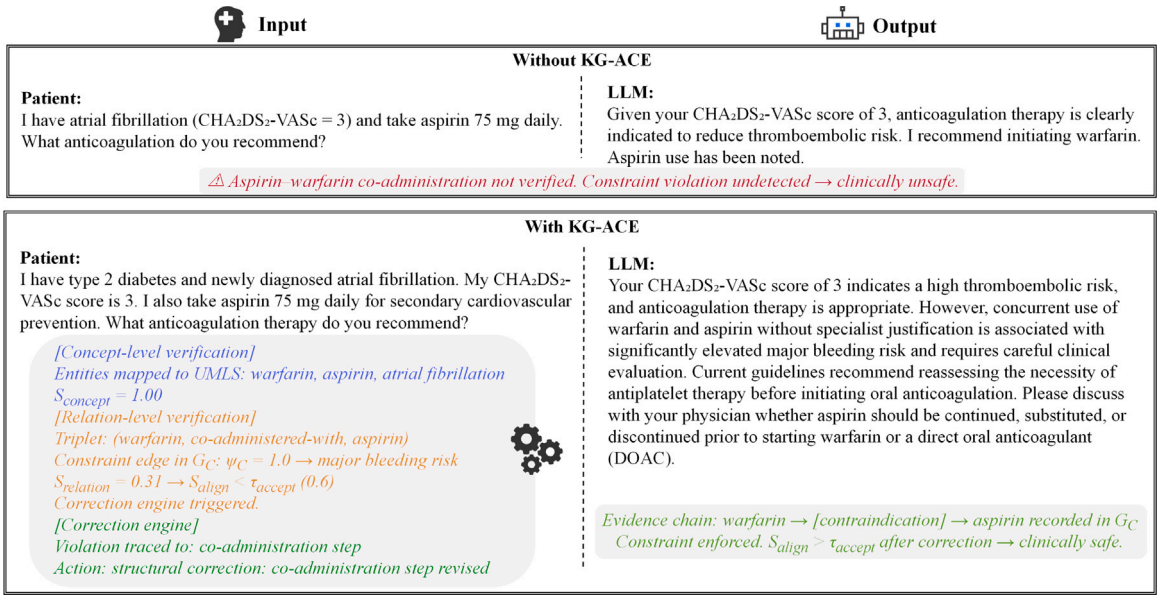


Fig. 1. Motivating example of the failure addressed by KG-ACE. Top: a standard LLM acknowledges concurrent aspirin use but does not verify the interaction during its intermediate reasoning, so a high-risk co-administration recommendation reaches the final answer. Bottom: KG-ACE verifies each reasoning step. The relation-level check identifies the high-severity warfarin–aspirin interaction encoded in $\mathcal{G}_C$, invokes the correction engine, and accepts the revision only when it raises S_{align} above τ_{accept}. The corrected output is therefore linked to an explicit and traceable evidence path.

training. These methods do not necessarily verify how that knowledge is used after generation. As a result, an answer may be correct while its intermediate reasoning remains incomplete or unsafe, and the supporting evidence may not be traceable to a specific medical source (Ji et al., 2023; Lee et al., 2023; Singhal et al., 2023b).

Motivation. Fig. 1 illustrates a process-level error. The model recommends warfarin for a patient with atrial fibrillation but fails to examine the concurrent use of aspirin, which creates a clinically important bleeding-risk interaction. The isolated recommendation may appear plausible; the problem arises in the omitted verification step. This example motivates three observations:

(M1) Final-answer evaluation can miss errors in intermediate reasoning. Drug interactions, conflicting clinical states, and incorrect temporal orderings may appear within a reasoning chain even when the final answer is superficially acceptable.

(M2) Linguistic fluency does not ensure logical completeness. A model may rely on statistical association and omit the intermediate conditions needed to justify a clinical inference.

(M3) General hallucination detection provides limited diagnostic detail. A single answer-level judgment does not identify whether an error originates from an unrecognized concept, an invalid relation, or a global logical conflict, and it does not indicate which part of the reasoning should be revised.

These observations motivate a post-generation framework that verifies the reasoning process at multiple levels and uses the verification result to guide targeted revision. The experimental evaluation is organized around four research questions (RQ1–RQ4) in Section 4; computational cost is examined in Appendix A.

Challenges. Three technical challenges follow from this motivation.

(C1) Reasoning consistency: no logic-level validation. Input-stage knowledge injection and answer-level checking do not determine whether each intermediate step conforms to the relevant medical knowledge structure. Internal contradictions can therefore remain undetected (Lee et al., 2023).

(C2) Reasoning completeness: association can replace clinical inference. Models may compress a multi-step pathophysiological process into an unsupported leap, omitting the conditions needed to connect evidence to a conclusion (Ji et al., 2023).

(C3) Reasoning traceability: evidence is not linked to individual steps. Clinical users need to know which knowledge source supports a recommendation. Current systems often provide no explicit mapping from a generated step to the entities, relations, or rules on which it depends (Singhal et al., 2023b).

Approach. KG-ACE addresses these challenges with a combined verification and correction pipeline. The Ontology Knowledge Graph ($\mathcal{G}_O$) standardizes medical concepts and semantic relations. The Consistency Knowledge Graph ($\mathcal{G}_C$) represents clinical constraints,

Table 1
Comparison of KG-ACE with closely related KG-LLM methods along four technical dimensions. ✓ : supported; × : not supported; ∼ : partial support.

Method	Post-gen. verification	Step-level error ID	Multi-gran. alignment	Benefit-driven correction
ToG (Sun et al., 2024)	×	×	×	×
MedGraphRAG (Wu et al., 2024)	∼	×	×	×
MGPrompt (Gong et al., 2026)	×	×	∼	×
OntologyGraphRAG (Ali et al., 2026)	∼	×	×	×
KG-ACE (ours)	✓	✓	✓	✓

including interactions, contraindications, causal relations, and temporal orderings. A three-level mechanism then evaluates concept validity, relation validity, and logical coherence. Finally, the correction engine revises only the affected reasoning content and retains a candidate revision only when its alignment score improves.

Contributions. KG-ACE shifts verification from the final answer to the generated reasoning process. It differs from existing work in three linked ways: it verifies reasoning after generation rather than only supplying knowledge before generation; it localizes errors at the concept, relation, and logic levels rather than returning a single answer-level verdict; and it accepts a correction only after measuring its effect on alignment. The knowledge graph therefore functions as an explicit verification specification rather than only as a retrieval source.

Table 1 compares KG-ACE with closely related methods that combine knowledge graphs and LLMs. ToG (Sun et al., 2024) searches a knowledge graph before answer generation, and MedGraphRAG (Wu et al., 2024) retrieves graph-grounded evidence. Neither method verifies the internal consistency of a completed reasoning chain. KG-ACE instead combines post-generation, step-level verification; alignment at the concept, relation, and logic levels; and benefit-driven correction. The framework is intended to complement, rather than replace, retrieval, prompting, and domain-specific training.

Specifically, this study makes three contributions:

- We develop a dual-layer medical knowledge graph that separates standardized medical concepts from clinical constraints, enabling the system to distinguish terminology errors from rule violations.
- We introduce a three-level verification mechanism that evaluates generated reasoning at the concept, relation, and logic levels and returns both a continuous alignment score and a structured diagnostic report.
- We design a benefit-driven correction engine that applies local, structural, or global revisions and retains a change only when the alignment improvement exceeds a specified threshold.

2. Related work

This section reviews four lines of research relevant to KG-ACE: medical knowledge augmentation, hallucination detection, explicit reasoning chains, and the integration of knowledge graphs with LLMs.

Medical knowledge augmentation. Retrieval-Augmented Generation (RAG) retrieves external evidence before generation and has become a common basis for medical knowledge augmentation (Lewis et al., 2020). DR.KNOWS retrieves patient-specific paths from medical knowledge graphs for diagnosis; MedCPT uses contrastive learning to improve matching between queries and documents; and KG-Rank re-ranks graph triples before they are supplied to the model (Jin et al., 2024; Xiong et al., 2024; Yang et al., 2024). KoSEL further reduces irrelevant knowledge by retrieving question-specific subgraphs (Zeng et al., 2025). These methods improve the evidence available at generation time, but they do not verify whether the generated intermediate steps use that evidence consistently. Related work on medical dialogue has improved clinical-context representation through speaker-aware and context-enhanced encoding (Li et al., 2025). KG-ACE addresses a different problem: it checks whether a completed reasoning process complies with explicit medical concepts and constraints.

Hallucination detection and mitigation. SelfCheckGPT detects factual uncertainty by comparing multiple samples from the same model, without requiring an external knowledge source (Manakul et al., 2023). Medical benchmarks such as Med-HALT show that reasoning errors remain a major source of failure, even when relevant medical knowledge is available (Umapathi et al., 2023). Knowledge-base fact checking can identify unsupported final claims, while ontology-aware GraphRAG constrains generation with formal semantic knowledge (Ali et al., 2026; Pal et al., 2023). Recent IPM research has also used evidence mining and multi-agent consensus to produce more transparent fact-verification decisions (Xiong et al., 2025). These approaches primarily judge claims or final answers. KG-ACE instead localizes violations within a domain-specific reasoning chain and uses the detected violation to select a correction scope.

Explicit reasoning chains. Chain-of-Thought prompting elicits intermediate steps, and Tree of Thoughts extends this idea by exploring and evaluating multiple candidate paths (Wei et al., 2022; Yao et al., 2023). In medical question answering, explicit reasoning can improve accuracy, but the generated steps may still lack verifiable support from medical knowledge sources (Liévin

et al., 2024; Savage et al., 2024). Making a reasoning chain visible is therefore not equivalent to establishing that the chain is clinically valid. KG-ACE treats explicit reasoning as an object to be checked, not as evidence of correctness by itself.

Knowledge graphs and LLMs. Research at the intersection of knowledge graphs and LLMs follows two broad directions: LLMs can assist knowledge-graph construction, and knowledge graphs can supply structured knowledge to LLMs (Pan et al., 2024). Examples include graph-neural-network integration for question answering and automatic construction of medical knowledge graphs from scientific literature (Christopoulou et al., 2023; Yasunaga et al., 2022). Recent IPM studies show that LLM-based graph construction remains sensitive to extraction and prompt design, while path-based models can improve the explainability of temporal knowledge-graph reasoning (Gajo & Barrón-Cedeño, 2025; Li & Wu, 2025). MGPrompt injects entity-, relation-, and subgraph-level knowledge to guide structured KGQA reasoning (Gong et al., 2026). These methods demonstrate the value of structured knowledge, but access to a knowledge graph does not guarantee that a generated reasoning chain follows the graph correctly.

Retrieval-based augmentation and constrained decoding likewise operate mainly before or during generation (Dathathri et al., 2020; Lu et al., 2022). They can improve grounding or control, but they do not directly test a completed reasoning chain at the concept, relation, and logic levels. KG-ACE fills this gap through post-generation verification and targeted correction. Its contribution is therefore complementary to knowledge retrieval, constrained generation, and path-based knowledge-graph reasoning.

3. Method

KG-ACE verifies a model-generated reasoning text after generation and revises it only when a detected inconsistency can be reduced. Fig. 2 shows the three components of the framework:

- **Dual-Layer Knowledge Graph Architecture.** The Ontology Knowledge Graph ($\mathcal{G}_O$) represents standardized medical concepts and semantic relations. The Consistency Knowledge Graph ($\mathcal{G}_C$) represents clinical constraints, including drug interactions, contraindications, causal relations, and temporal orderings (Section 3.1).
- **Three-Level Alignment Verification Mechanism.** The mechanism checks concept validity, relation validity, and logical coherence and combines the three results in the alignment score S_{align} (Section 3.2).
- **Intelligent Correction Engine.** The engine applies a local, structural, or global revision according to the diagnostic report. A candidate revision is retained only when it improves the alignment score by more than the minimum improvement threshold (Section 3.3).

The components form a closed loop. The two knowledge graphs define the concepts and constraints to be checked; the verification mechanism identifies and scores violations; and the correction engine revises the affected reasoning before the output is evaluated again. Throughout the paper, the capitalized terms *Ontology Knowledge Graph* and *Consistency Knowledge Graph* refer specifically to $\mathcal{G}_O$ and $\mathcal{G}_C$, respectively.

3.1. Dual-layer knowledge graph architecture

This subsection describes five elements of the architecture: (*i*) the Ontology Knowledge Graph, (*ii*) the Consistency Knowledge Graph, (*iii*) medical entity standardization, (*iv*) constraint-rule representation, and (*v*) graph construction.

(*i*) Ontology Knowledge Graph.

The Ontology Knowledge Graph is a directed, labeled graph in which nodes represent standardized medical concepts and edges represent hierarchical or semantic relations. Formally, $\mathcal{G}_O = (V_O, E_O, \phi_O)$, where V_O is the concept set, E_O is the relation set, and ϕ_O maps each concept to a category in $\mathcal{C}$. For example, the graph may connect *insulin* to *antidiabetic drug* through an *is-a* relation. This representation supports concept normalization and semantic-type checking without treating a treatment relation as an ontological edge. Each extracted medical mention is matched to a standardized concept through a synonym index or semantic similarity. Mapping variants of the same term to one concept reduces errors caused by inconsistent terminology and provides stable entity identifiers for subsequent relation and logic checks.

(*ii*) Consistency Knowledge Graph.

The Consistency Knowledge Graph encodes rules that determine whether relations in a reasoning chain are clinically compatible. Formally, $\mathcal{G}_C = (V_C, E_C, \psi_C)$, where V_C contains the entities involved in clinical rules, E_C contains directed constraint relations, and $\psi_C : E_C \rightarrow [0,1]$ records the confidence or severity assigned to each rule. The relation types include drug interactions, contraindications, causal relations, mutual exclusions, and temporal orderings. For example, concurrent warfarin and aspirin use can be represented by a high-severity interaction edge because the combination may substantially increase bleeding risk. A non-selective beta-blocker and asthma can likewise be connected by a high-severity contraindication edge, whereas a selective β_1 -blocker may receive a lower rule weight when the underlying guideline permits cautious use in specific patients. The graph can also encode order constraints such as *biopsy precedes chemotherapy* and causal relations such as *chronic hypertension can contribute to heart failure*. During verification, KG-ACE queries these edges to determine whether a generated relation is supported, conditionally acceptable, or prohibited. The edge weight preserves differences in rule strength rather than reducing all constraints to a binary decision.

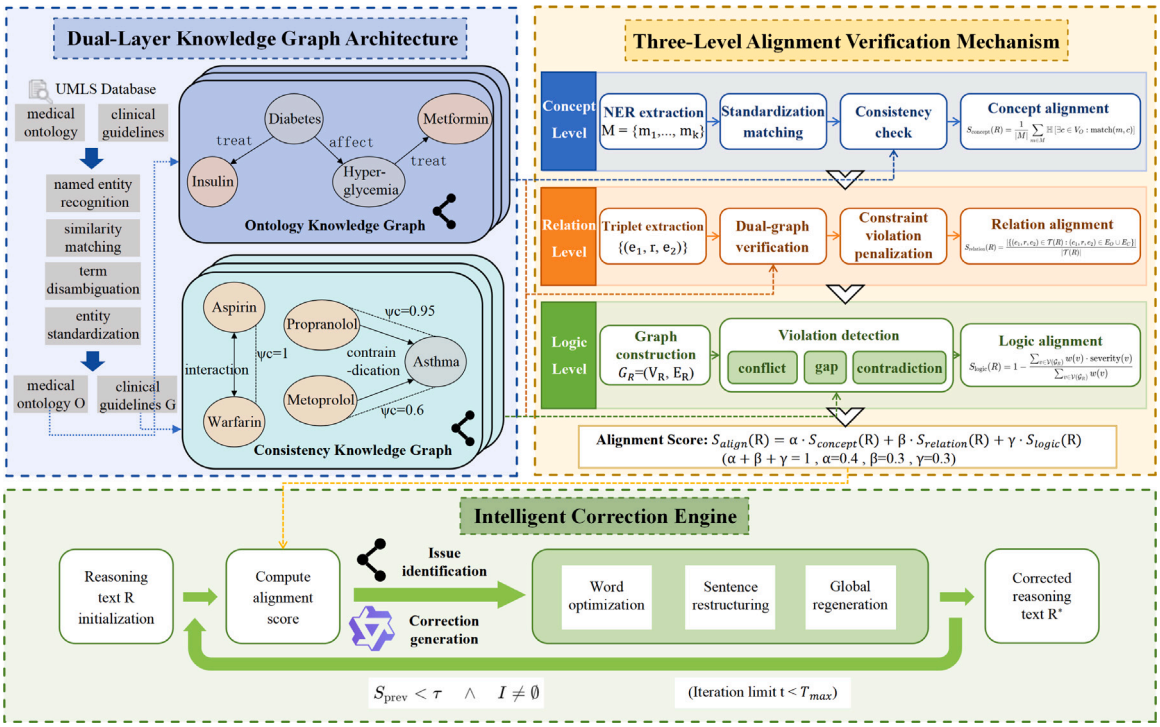


Fig. 2. Overall architecture of KG-ACE. (1) The dual-layer architecture constructs the Ontology Knowledge Graph and the Consistency Knowledge Graph from medical ontologies and clinical guidelines. (2) The verification mechanism evaluates the generated reasoning at the concept, relation, and logic levels and combines the partial scores in S_{align}. (3) The correction engine applies local, structural, or global revisions and retains a revision only when it improves the alignment score.

(*iii*) Medical Entity Standardization.

KG-ACE first applies named entity recognition (NER) to extract medical mentions from the reasoning text. Let $\hat{c} = \arg\max_{c \in V_O} \text{sim}(m, c)$ be the concept most similar to mention m. Standardization is then defined as

$$\text{Standardize}(m) = \begin{cases} \hat{c}, & \text{sim}(m, \hat{c}) > \tau_s, \\ \varnothing, & \text{otherwise}, \end{cases} \tag{1}$$

where $\text{sim}(m, c)$ is the semantic similarity between mention m and concept c, and τ_s is the matching threshold. A mention is therefore linked only when its best candidate exceeds the threshold; otherwise, it remains unresolved and is reported to the verification mechanism.

(*iv*) Representation of Constraint Rules.

Clinical constraints are represented as first-order logic predicates when a rule can be expressed deterministically. For example, the contraindication of beta-blockers for asthmatic patients is expressed as:

$$\forall d, p : \text{IsNonselectiveBetaBlocker}(d) \wedge \text{HasAsthma}(p) \Rightarrow \neg\text{Prescribe}(d, p), \tag{2}$$

where d denotes a drug, p denotes a patient, $\text{IsNonselectiveBetaBlocker}(d)$ indicates that drug d is a non-selective beta-blocker, $\text{HasAsthma}(p)$ indicates that patient p has a history of asthma, and $\neg\text{Prescribe}(d, p)$ indicates that the drug should not be prescribed to this patient. KG-ACE extracts such rules from clinical guidelines and converts them into weighted graph edges for subsequent verification.

(*v*) Construction of the Dual-Layer Knowledge Graph.

The two graphs are constructed from complementary sources: medical ontologies supply standardized concepts and semantic relations, whereas clinical guidelines supply constraint rules. Algorithm 1 summarizes the construction procedure.

Algorithm 1 Dual-Layer Knowledge Graph Construction

1: **Input:** Medical ontology O, Clinical guidelines G
2: **Output:** Ontology Knowledge Graph $\mathcal{G}_O$, Consistency Knowledge Graph $\mathcal{G}_C$
3: Initialize empty graphs $\mathcal{G}_O \leftarrow$ EmptyGraph(), $\mathcal{G}_C \leftarrow$ EmptyGraph()
4: **for** each concept $c \in O$ **do**
5: Add node $(c.id, c.name, c.category)$ to $\mathcal{G}_O$
6: **for** each synonym s in $c.synonyms$ **do**
7: Update synonym_index[s] $\leftarrow c.id$
8: **end for**
9: **end for**
10: **for** each clinical guideline $g \in G$ **do**
11: Parse constraints C from guideline g
12: **for** each constraint $(e_1, relation, e_2, severity) \in C$ **do**
13: Add edge $(e_1, relation, e_2)$ with weight $severity$ to $\mathcal{G}_C$
14: **end for**
15: **end for**
16: **return** $\mathcal{G}_O, \mathcal{G}_C$

The inputs are a medical ontology O and a set of clinical guidelines G. The ontology provides standardized medical concepts to be mapped into $\mathcal{G}_O$, while the guidelines supply constraint rules such as contraindications and drug interactions, which are converted into edges and nodes in $\mathcal{G}_C$. The outputs are the fully constructed $\mathcal{G}_O$ and $\mathcal{G}_C$. The algorithm begins by initializing two empty graphs (line 3). $\mathcal{G}_O$ is used to store medical concepts and their hierarchical relations, while $\mathcal{G}_C$ stores constraint relations extracted from clinical guidelines. In lines 4–8, each concept c in the ontology is added as a node to $\mathcal{G}_O$, and its synonyms are indexed to the corresponding concept ID to support efficient matching during inference. In lines 9–14, the algorithm constructs $\mathcal{G}_C$ by parsing constraint rules from each guideline g. Each rule is represented as a quadruple $(e_1, relation, e_2, severity)$, where e_1 and e_2 are medical entities, $relation$ is the type of constraint (e.g., contraindication or interaction), and $severity$ reflects the strength of the constraint. These rules are added as weighted edges to $\mathcal{G}_C$. Line 15 returns the completed $\mathcal{G}_O$ and $\mathcal{G}_C$, which serve as the structured knowledge foundation for medical reasoning.

3.2. Three-level alignment verification mechanism

The verification mechanism processes the generated reasoning text in the order concept → relation → logic. Concept-level verification first links medical mentions to $\mathcal{G}_O$. Relation-level verification then extracts triples from the normalized text and compares them with $E_O \cup E_C$. Logic-level verification constructs a reasoning graph and checks global causal, temporal, and contradiction constraints. Fig. 3 summarizes this pipeline.

The ordered decomposition reflects a dependency in the verification task. A relation cannot be evaluated reliably until its entities have been normalized, and global logic cannot be evaluated reliably until the relevant relations have been identified. Keeping the three error classes separate also makes the diagnostic report actionable: the report states the verification level, affected entities or propositions, violation type, and severity. Continuous partial scores distinguish incomplete support from a complete mismatch and guide the correction engine more precisely than a single binary verdict. The ablation study and sensitivity analysis evaluate the contribution and robustness of this decomposition.

Given a model-generated reasoning text R, the overall alignment score is defined as a weighted linear combination of scores from the three verification levels:

$$S_{\text{align}}(R) = \alpha \cdot S_{\text{concept}}(R) + \beta \cdot S_{\text{relation}}(R) + \gamma \cdot S_{\text{logic}}(R), \tag{3}$$

where the weights satisfy the normalization constraint $\alpha + \beta + \gamma = 1$. The weights reflect both the dependency between levels and validation performance. Concept verification receives the largest weight because relation and logic checks depend on correct entity normalization. Relation and logic verification receive equal weights: the former checks explicit graph constraints, whereas the latter checks broader structural coherence. A grid search on held-out validation data used $\alpha \in \{0.3, 0.4, 0.5\}$ and $\beta \in \{0.2, 0.3, 0.4\}$, with $\gamma = 1 - \alpha - \beta$. The best configuration was $(\alpha, \beta, \gamma) = (0.4, 0.3, 0.3)$, which is used in all main experiments (Section 4.6).

A reasoning output passes verification when $S_{\text{align}}(R) \geq \tau_{\text{accept}}$. The default threshold is $\tau_{\text{accept}} = 0.6$, selected on validation data to balance the rejection of poorly aligned outputs against unnecessary correction of acceptable outputs.

(*i*) Concept-Level Verification.

Concept-level verification focuses on the correctness and coverage of medical entities in the reasoning text. Named entity recognition is applied to extract a set of medical mentions $M = \{m_1, m_2, \ldots, m_k\}$ from reasoning text R, which may include disease names, drug components, and clinical indicators. Each mention is then matched against the standard concept set V_O in the Ontology Knowledge Graph $\mathcal{G}_O$. The concept alignment score is the proportion of extracted mentions that can be standardized:

$$S_{\text{concept}}(R) = \frac{1}{\max(|M|, 1)} \sum_{m \in M} \mathbb{1}\left[\exists c \in V_O : \text{match}(m, c)\right], \tag{4}$$

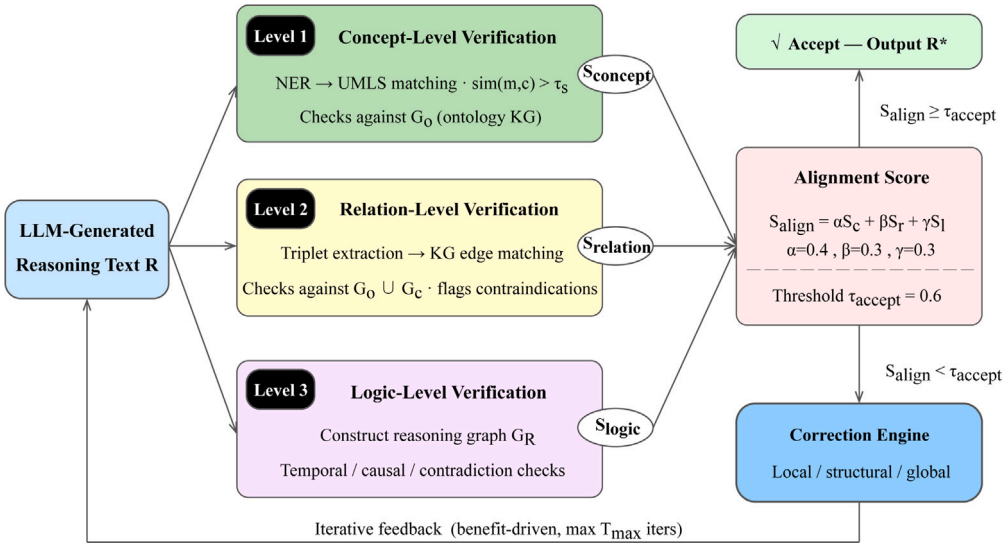


Fig. 3. The three-level alignment verification mechanism of KG-ACE. @Reasoning text R is processed sequentially through concept-level, relation-level, and logic-level verification, each querying the dual-layer knowledge graph ($\mathcal{G}_O$ and $\mathcal{G}_C$) to produce a partial score. The three scores are combined into a weighted alignment score S_{align}. Reasoning that meets the acceptance threshold τ_{accept} is directly output; otherwise, the intelligent correction engine applies iterative, benefit-driven corrections and resubmits the revised text for re-evaluation.

where the indicator function $\mathbb{1}[\cdot]$ equals 1 when the condition is satisfied and 0 otherwise. The matching function $\text{match}(m, c)$ combines semantic similarity with synonym lookup:

$$\text{match}(m, c) \iff \left(\text{sim}(m, c) > \tau_s\right) \vee \left(m \in \sigma^{-1}(c)\right). \tag{5}$$

Here $\text{sim}(\cdot)$ denotes semantic embedding similarity computed using a pretrained medical language model, τ_s is the similarity threshold (default: 0.85), and $\sigma^{-1}(c)$ is the set of synonyms associated with concept c. This dual-path matching strategy handles the diverse ways in which medical terms are expressed, supporting both exact and approximate semantic matching.

Concept-level verification goes beyond checking whether entities appear in the knowledge graph. It also enforces semantic type consistency, so that valid concepts are not accepted when they are used in clinically incoherent roles. For example, if a reasoning path contains a type mismatch such as *using diabetes to treat hypertension*, the alignment score for that segment is penalized even though both *diabetes* and *hypertension* are valid concepts.

(*ii*) Relation-Level Verification.

Building on concept-level verification, relation-level verification examines whether the relations between medical entities in the reasoning text conform to domain knowledge constraints. A relation extraction module identifies triplet structures $\{(e_1, r, e_2)\}$ from the reasoning path, where e_1 and e_2 are standardized medical entities and r is a relation type. The relation alignment score is defined as the proportion of verified relations among all extracted relations:

$$S_{\text{relation}}(R) = \frac{\left|\left\{(e_1, r, e_2) \in \mathcal{T}(R) : (e_1, r, e_2) \in E_O \cup E_C\right\}\right|}{\max(|\mathcal{T}(R)|, 1)}, \tag{6}$$

where $\mathcal{T}(R)$ denotes the set of triplets extracted from reasoning text R, and E_O and E_C are the edge sets of the Ontology Knowledge Graph and Consistency Knowledge Graph, respectively. If no relation can be extracted, the numerator is zero and the score is set to zero, indicating that relation-level support has not been established.

For constraint relations defined in $\mathcal{G}_C$, the system performs bidirectional verification. If the reasoning path contains a relation combination that is flagged as prohibited, such as *prescribing a beta-blocker to an asthmatic patient*, the relation-level score is substantially reduced even if the relation is syntactically valid.

(*iii*) Logic-Level Verification.

Logic-level verification examines the internal coherence and causal-temporal validity of the reasoning chain. This level requires the broadest context of the three checks. Unlike concept-level and relation-level verification, which focus on local checks, logic-level verification requires a global consistency analysis of the entire reasoning path, identifying structural defects such as logical leaps, circular reasoning, and temporal inversions.

The reasoning text is structured as a reasoning graph $\mathcal{G}_R = (V_R, E_R)$, where nodes V_R represent key propositions in the reasoning steps and edges E_R represent inferential relations between them. A violation detection function $\mathcal{V} : \mathcal{G}_R \to 2^{\mathcal{I}}$ maps the reasoning

graph to a set of detected violations, where $\mathcal{I}$ is the set of all possible violation types. The logic alignment score is computed as the inverse of weighted violation severity:

$$S_{\text{logic}}(R) = 1 - \frac{\sum_{v \in \mathcal{V}(\mathcal{G}_R)} w(v) \cdot \text{severity}(v)}{\max\left(\sum_{v \in \mathcal{V}(\mathcal{G}_R)} w(v), 1\right)}, \tag{7}$$

where $\text{severity}(v) \in [0, 1]$ quantifies the severity of violation v and $w(v)$ is its associated weight. Severity values follow a hierarchical principle: logical errors that directly threaten patient safety, such as prescribing a contraindicated drug, are assigned high severity values close to 1, while minor expression issues receive lower values.

Violation detection covers multiple dimensions. Temporal consistency checks verify that diagnosis precedes treatment and that symptoms precede diagnosis, in accordance with standard clinical workflows. Causal chain completeness checks analyze the connectivity of the reasoning graph to identify missing intermediate steps. Contradiction detection uses mutual exclusion relations in $\mathcal{G}_C$ to identify self-contradictions within the reasoning path. Algorithm 2 describes the logic-level verification procedure in detail.

Algorithm 2 Logic-Level Alignment Verification

1: **Input:** Reasoning text R, Consistency Knowledge Graph $\mathcal{G}_C$, Violation weight table W
2: **Output:** Logic alignment score S_{logic}
3: Initialize reasoning graph $\mathcal{G}_R \leftarrow \text{ConstructReasoningGraph}(R)$
4: Initialize violation set $\mathcal{I} \leftarrow \emptyset$
5: **for** each node pair $(v_i, v_j) \in V_R \times V_R$ **do**
6: **if** $\text{HasTemporalConflict}(v_i, v_j)$ **then**
7: Add $(v_i, v_j, \text{“temporal”})$ to $\mathcal{I}$
8: **end if**
9: **end for**
10: **for** each path $p = (v_1 \rightarrow v_2 \rightarrow \cdots \rightarrow v_n)$ in $\mathcal{G}_R$ **do**
11: gaps $\leftarrow \text{DetectCausalGaps}(p, \mathcal{G}_C)$
12: Add gaps to $\mathcal{I}$
13: **end for**
14: **for** each node $v \in V_R$ **do**
15: contradictions $\leftarrow \text{FindContradictions}(v, \mathcal{G}_C)$
16: Add contradictions to $\mathcal{I}$
17: **end for**
18: total_weight $\leftarrow \sum_{i \in \mathcal{I}} W[\text{type}(i)]$
19: penalty $\leftarrow \sum_{i \in \mathcal{I}} W[\text{type}(i)] \cdot \text{severity}(i)$
20: $S_{\text{logic}} \leftarrow 1 - \frac{\text{penalty}}{\max(\text{total_weight}, 1)}$
21: **return** S_{logic}

The inputs are the reasoning text R, the Consistency Knowledge Graph $\mathcal{G}_C$, and a violation weight table W. The two helper functions in Algorithm 2 are specified as follows. **DetectCausalGaps**$(p, \mathcal{G}_C)$ receives a reasoning path $p = (v_1 \rightarrow v_2 \rightarrow \cdots \rightarrow v_n)$ and checks whether each consecutive pair (v_i, v_{i+1}) is connected by a valid causal or temporal edge in $\mathcal{G}_C$. A gap is recorded for any pair for which no such edge exists and for which the conceptual distance (measured as the shortest path length in $\mathcal{G}_O$) exceeds a threshold $\delta_{\text{gap}} = 2$, indicating a multi-hop inferential leap that has been collapsed into a single step. The value $\delta_{\text{gap}} = 2$ was selected based on the semantic structure of $\mathcal{G}_O$: a shortest-path distance of 1 corresponds to directly adjacent ontological concepts (e.g., *hypertension* → *cardiovascular disease*), which may legitimately be condensed in clinical prose; a distance of 2 or greater indicates that at least one intermediate concept has been skipped, constituting an inferential leap that warrants flagging. To assess the sensitivity of this threshold, we varied $\delta_{\text{gap}} \in \{1, 2, 3\}$ on the MedQA validation split and observed that $\delta_{\text{gap}} = 1$ produced excessive false-positive gap detections (reducing S_{logic} even for well-formed chains), while $\delta_{\text{gap}} = 3$ missed a non-trivial proportion of genuine multi-hop leaps confirmed by manual inspection. The value of 2 achieved the best balance between precision and recall of gap detection on this split and is adopted as the default. **FindContradictions**$(v, \mathcal{G}_C)$ retrieves all outgoing edges of the proposition node v in the reasoning graph $\mathcal{G}_R$ and checks each corresponding entity pair against the mutual-exclusion relations encoded in E_C. A contradiction is recorded when two propositions asserted within the same reasoning chain are connected by a `mutually_exclusive` or `contraindicated` edge in $\mathcal{G}_C$. Temporal relations are parsed from free text using a rule-based tagger that identifies markers such as “prior to”, “following”, “after”, and “before”, supplemented by the `BEFORE/AFTER` relation types available in the UMLS Semantic Network.

The algorithm begins by constructing the reasoning graph $\mathcal{G}_R$ from R and initializing an empty violation set $\mathcal{I}$ (lines 3–4). In lines 5–8, temporal conflicts between node pairs are detected; any conflicting pair is added to $\mathcal{I}$. In lines 9–12, causal gaps along each reasoning path are identified using $\text{DetectCausalGaps}(p, \mathcal{G}_C)$ and added to $\mathcal{I}$. In lines 13–16, contradictions at each node are detected against $\mathcal{G}_C$ and recorded. Lines 17–19 compute the total weighted penalty and derive the final logic alignment score S_{logic}. By combining temporal conflict detection, causal gap identification, and contradiction checking with weighted

penalization, the algorithm produces a quantitative measure of logical consistency that guides subsequent correction. When the overall alignment score S_{align} falls below the acceptance threshold τ_{accept}, the verification mechanism outputs a structured diagnostic report enumerating the violation types, involved entities, severity ratings, and the verification level at which each issue was detected. This report is passed directly to the intelligent correction engine, which uses it to select the appropriate correction mode and scope for each identified issue, as described in the following subsection.

3.3. Intelligent correction engine

The correction engine receives the structured diagnostic report produced by the verification mechanism. It applies a candidate revision, recomputes the alignment score, and retains the revision only when the improvement exceeds ϵ. The loop ends when the reasoning reaches the acceptance threshold, no issue remains, no candidate produces sufficient improvement, or the maximum number of iterations is reached. Formally,

$$R^{(t+1)} = \text{Correct}(R^{(t)}, \mathcal{I}^{(t)}), \tag{8}$$

where $\mathcal{I}^{(t)}$ is the issue set at iteration t. A *local correction* replaces or normalizes an affected term. A *structural correction* revises a flagged step and the downstream steps that depend on it. A *global correction* regenerates the complete reasoning chain when the diagnostic report identifies a system-wide temporal, causal, or contradictory structure.

To resolve the ambiguity in how corrections are operationally executed, we specify the correction prompt generation procedure in Algorithm 3. Each violation type detected during verification triggers a corresponding correction mode and a structured prompt presented to the LLM.

Algorithm 3 Correction Prompt Generation

Input: Current reasoning text R^*, Issue set $\mathcal{I}^{(t)}$, KG subgraph $\mathcal{G}_{\text{sub}}$ consisting of entities and edges in $\mathcal{V}(\mathcal{G}_R)$ involved in violations
Output: Candidate correction set C
$C \leftarrow \emptyset$
for each issue $i \in \mathcal{I}^{(t)}$ **do**
  if type(i) = `concept_mismatch` **then**
    prompt $\leftarrow$ LocalPrompt(i.entity, i.umls_standard, R^*)
  else if type(i) = `constraint_violation` **then**
    prompt $\leftarrow$ StructuralPrompt($i.e_1$, $i.r$, $i.e_2$, $i.\psi_C$, i.step_id, R^*)
  else if type(i) $\in$ {`temporal_conflict`, `causal_gap`, `contradiction`} **then**
    prompt $\leftarrow$ GlobalPrompt($\mathcal{V}(\mathcal{G}_R)$, R^*)
  end if
  $c \leftarrow$ LLM(prompt)
  Add c to C
end for
return C

The three prompt templates follow a structured format comprising a system-role declaration, a runtime-populated context block, and a constrained instruction block. LocalPrompt instructs the LLM to replace a detected non-standard term with its UMLS-normalized equivalent while modifying only the affected phrase. StructuralPrompt instructs the LLM to revise a flagged step and its logically dependent downstream steps to eliminate a specific constraint violation without altering unaffected steps. GlobalPrompt instructs the LLM to regenerate the complete reasoning chain such that all detected violations are resolved and clinical logic is internally consistent. Full prompt templates, including runtime variable substitution procedures, are provided verbatim in Appendix C.

The inputs to Algorithm 4 are the reasoning text R, the maximum number of iterations $T_{\max}$, an acceptance threshold τ_{accept}, and a minimum improvement threshold ϵ. The outputs are the revised reasoning text R^* and the final alignment score S_{prev}. The algorithm initializes R^* and computes the initial alignment score S_{prev} (lines 3–4). In each iteration, issues in the current reasoning text are identified (line 6). If no issues remain or the score already meets τ_{accept}, the loop terminates (lines 7–9). Otherwise, candidate corrections are generated (line 10), and each correction c is applied to produce a temporary text R_{temp} (line 12). The alignment score of R_{temp} is computed (line 13), and the correction is accepted only if the increase in score exceeds ϵ (lines 14–16). This benefit-driven acceptance criterion prevents unnecessary modifications that could reduce fluency without improving alignment. The process repeats until $T_{\max}$ is reached or the reasoning meets the acceptance standard. The final corrected text R^* and its alignment score S_{prev} are returned (line 20). For Algorithm 4, all core experiments use a maximum of $T_{\max} = 5$ correction iterations and a minimum accepted alignment gain of $\epsilon = 0.01$. We selected $T_{\max} = 5$ from a validation sweep over $\{3, 5, 7\}$: $T_{\max} = 3$ occasionally stopped before multi-step constraint violations were resolved, whereas $T_{\max} = 7$ yielded negligible additional accuracy at the cost of higher latency. The observed mean of 2.3 correction iterations per query in Appendix A shows that the loop usually terminates well before reaching this upper bound. Through this iterative process, the intelligent correction engine dynamically refines each component of the reasoning chain, progressively improving logical consistency and ensuring that the final output conforms to established medical knowledge and clinical standards.

Algorithm 4 Iterative Correction and Feedback Loop

Input: Reasoning text R, Maximum iterations $T_{\max}$, Acceptance threshold τ_{accept}, Improvement threshold ϵ
Output: Corrected reasoning text R^*, Final alignment score S_{prev}
$t \leftarrow 0$, $R^* \leftarrow R$
$S_{\text{prev}} \leftarrow \text{Align}(R^*)$
while $t < T_{\max}$ **do**
  $\mathcal{I}^{(t)} \leftarrow \text{IdentifyIssues}(R^*)$
  if $\mathcal{I}^{(t)} = \emptyset$ **or** $S_{\text{prev}} \geq \tau_{\text{accept}}$ **then**
    break
  end if
  $C \leftarrow \text{GenerateCorrections}(R^*, \mathcal{I}^{(t)})$
  for each correction $c \in C$ **do**
    $R_{\text{temp}} \leftarrow \text{Apply}(c, R^*)$
    $S_{\text{temp}} \leftarrow \text{Align}(R_{\text{temp}})$
    if $S_{\text{temp}} > S_{\text{prev}} + \epsilon$ **then**
      $R^* \leftarrow R_{\text{temp}}$
      $S_{\text{prev}} \leftarrow S_{\text{temp}}$
    end if
  end for
  $t \leftarrow t + 1$
end while
return R^*, S_{prev}

4. Experiments

The experiments address four research questions:

- **RQ1:** How does KG-ACE compare with strong prompting, knowledge-graph, retrieval, and domain-pretrained baselines across medical reasoning tasks?
- **RQ2:** How much does each component contribute to performance?
- **RQ3:** How sensitive is KG-ACE to its main hyperparameters?
- **RQ4:** How does KG-ACE identify and revise an unsafe relation in an illustrative patient scenario?

The base model is Qwen2.5-7B-Instruct (Qwen Team, 2024). The Ontology Knowledge Graph is constructed from UMLS (Bodenreider, 2004), and the Consistency Knowledge Graph is constructed from the clinical-guideline database used in this study. The model and graph index are hosted on the same device, so graph queries do not require network communication. To make the evaluation computationally comparable across datasets, we sample at most 2000 test instances from each dataset by stratified random sampling. MedQA and GenMedGPT-5k contain fewer than 2000 test instances, so their complete test splits are used. This sampling protocol does not establish clinical population representativeness; Section 5.3 discusses this limitation. All experiments use random seed 42, and each reported result is the mean of three independent runs.

4.1. Datasets

We use four public datasets covering medical multiple-choice question answering, open-ended medical dialogue, and differential diagnosis. Table 2 summarizes the datasets and evaluation splits.

Medical multiple-choice question answering. MedQA (Jin et al., 2021) is based on USMLE questions, comprising 10,200 training and 1273 test questions with 4 to 5 answer options, and evaluates complex multi-step clinical reasoning. MedMCQA (Pal et al., 2022) contains over 194,000 questions from AIIMS and NEET-PG, covering 21 medical subjects and approximately 2400 health topics, and includes questions that require multi-step medical reasoning. Both datasets provide well-defined ground-truth answers, are evaluated using accuracy, and rank among the most widely used benchmarks in medical LLM research (Pal et al., 2024).

Doctor-patient dialogue. GenMedGPT-5k (Eyuboglu et al., 2023) consists of 5000 multi-turn medical dialogues simulated by GPT-3.5, covering symptom description, diagnostic reasoning, and treatment recommendations. Although both the reference dialogues and KG-ACE's base model are LLM-generated, the risk of circular evaluation is mitigated by two factors: the reference texts were produced by GPT-3.5 under a constrained simulation protocol distinct from Qwen2.5-7B's training distribution, and the primary evaluator is GPT-4 scoring with explicit medical correctness criteria and reference reasoning chains as anchors rather than surface similarity to the GPT-3.5 output. Nevertheless, the GenMedGPT-5k results should be interpreted as indicative rather than definitive, and we acknowledge this as a threat to external validity in Section 5.3. Since the reference answers in this dataset are generated

Table 2
Summary of evaluation datasets. MCQ: multiple-choice question answering; DDx: differential diagnosis. †Full test set used (fewer than 2000 examples available). Stratified random sampling with seed 42 applied to all other datasets.

Dataset	Task	Train	Test	Eval	Metrics
MedQA	MCQ	10,200	1273	1273†	Accuracy, RCS
MedMCQA	MCQ	187,599	6150	2000	Accuracy, RCS
GenMedGPT-5k	Dialogue	4000	1000	1000†	BS-F1, LLM-Score
DDXPlus	DDx	1,008,552	110,923	2000	Acc., Macro-F1, RCS

texts, standard string-matching metrics are not suitable for evaluating model outputs. We therefore evaluate this dataset with complementary reference-based and LLM-based metrics.

Differential diagnosis. DDXPlus (Fansi Tchango et al., 2022) is a large-scale synthetic dataset containing over one million patient cases. Each case provides a complete differential diagnosis list across 49 pathologies, a confirmed diagnosis, and detailed symptom and antecedent information. It is well suited for evaluating the coverage and logical soundness of reasoning paths under multi-symptom inputs. Given its large test split of 110,923 cases, 2000 samples are drawn via stratified random sampling for evaluation.

4.2. Evaluation metrics

The evaluation uses complementary metrics matched to the output format of each task and follows recent work on medical reasoning assessment (Chen et al., 2025; Zhou et al., 2026).

Closed-form question answering (MedQA, MedMCQA, DDXPlus). Accuracy is used as the primary metric, directly measuring the proportion of correct answers against ground truth. We additionally report the **Reasoning Consistency Score (RCS)** to measure the sampling stability of the generated reasoning structure. For each question, five independent samples are drawn, and the pairwise Jaccard overlap of nodes and edges between the resulting reasoning graphs is computed. The mean overlap across all pairs serves as the RCS, with higher values indicating more stable reasoning. This design is consistent with the multi-path sampling approach used in existing work to assess reasoning stability (Wang et al., 2023). It should be noted that RCS measures *reasoning stability*, defined as the reproducibility of the reasoning path structure under independent sampling, rather than reasoning correctness. A model may achieve high RCS by consistently reproducing the same incorrect intermediate steps, so RCS is always interpreted jointly with accuracy and other correctness metrics rather than as a standalone quality measure. Accordingly, high RCS combined with high accuracy indicates that the model reliably converges on correct reasoning paths, which is precisely the target behavior that KG-ACE's verification mechanism is designed to promote. The limitation that RCS has not been validated against expert clinical judgment of reasoning quality is acknowledged in Section 5.3. For DDXPlus, **Macro-F1** over the differential diagnosis list is additionally reported to assess coverage completeness: per-case Precision, Recall, and F1 are computed by matching model-output diagnoses against the ground-truth differential diagnosis set, and the mean F1 across all cases is reported.

Open-ended dialogue (GenMedGPT-5k). Since reference texts are available for this task, two metrics are used.

(1) **BERTScore** (Zhang et al., 2020): semantic similarity between the generated response and the reference text is computed using contextual embeddings from a pretrained language model, with the F1 score (BS-F1) as the reported value. Given that BERTScore has known limitations in correlating with expert judgment on medical texts (Zhou et al., 2026), BioMedBERT (Gu et al., 2020) is used as the encoder to improve domain adaptation. BERTScore is treated as a reference baseline for semantic similarity rather than a standalone quality measure.

(2) **GPT-4 Scoring (LLM-as-Judge)** (Zheng et al., 2023): GPT-4-turbo is used as the evaluator. Each response is scored from 0 to 1 on five dimensions: relevance, accuracy, completeness, clarity, and conciseness, with reference dialogue responses and medical reasoning chains used as anchors. The mean score across dimensions is reported as the LLM-Score. According to a 2025 empirical study, LLM scoring with reference reasoning chains achieves a Pearson correlation of 0.87 with expert evaluation, substantially outperforming reference-free LLM scoring and text similarity metrics such as BERTScore (Zhou et al., 2026). GPT-4 scoring is therefore used as the primary quality measure for open-ended tasks, with BERTScore serving as a supplementary reference.

4.3. Baseline methods

The comparison includes six general baselines and two Meditron-7B settings. Direct, CoT, and Self-Consistency represent single-pass, explicit-chain, and multi-sample prompting. MindMap, ToG, and MedGraphRAG represent input-stage graph prompting,

Table 3
Comparison of all methods on four datasets. Acc. denotes accuracy (%), RCS denotes the Reasoning Consistency Score, BS-F1 denotes the BERTScore F1 based on BioMedBERT, LLM-Score denotes the mean GPT-4 score across five dimensions, and Macro-F1 denotes the macro-averaged F1 over the differential diagnosis list on DDXPlus. **Bold** indicates the best result and underline indicates the second best. All results are averages over three independent runs. Meditron-7B results are included to assess additive value beyond domain-specific pretraining.

Method	MedQA		MedMCQA		GenMedGPT-5k		DDXPlus		
	Acc.	RCS	Acc.	RCS	BS-F1	LLM-Score	Acc.	RCS	Macro-F1
Direct	53.4	0.601	54.8	0.587	0.827	0.628	58.7	0.574	0.301
CoT (Wei et al., 2022)	58.9	0.643	60.3	0.629	0.841	0.672	64.2	0.619	0.334
SC (Wang et al., 2023)	61.7	0.688	62.9	0.671	0.845	0.691	66.8	0.641	0.352
MindMap (Wen et al., 2024)	63.4	0.697	64.7	0.682	0.856	0.718	68.5	0.653	0.371
ToG (Sun et al., 2024)	65.1	0.712	66.3	0.698	0.849	0.724	70.3	0.668	0.389
MedGraphRAG (Wu et al., 2024)	64.2	0.703	65.5	0.689	0.852	0.716	69.4	0.659	0.381
Meditron-7B (Direct) (Chen et al., 2023)	59.3	0.631	61.2	0.622	0.836	0.649	65.1	0.614	0.341
Meditron-7B + CoT (Chen et al., 2023)	66.8	0.724	67.9	0.711	0.851	0.703	69.7	0.651	0.374
KG-ACE	**69.8**	**0.774**	**70.9**	**0.761**	**0.868**	**0.791**	**75.1**	**0.739**	**0.463**
Δ(vs. ToG)	+4.7	+0.062	+4.6	+0.063	+0.012	+0.067	+4.8	+0.071	+0.074
Δ(vs. Meditron+CoT)	+3.0	+0.050	+3.0	+0.050	+0.017	+0.088	+5.4	+0.088	+0.089

iterative graph traversal, and graph-based retrieval. Meditron-7B is evaluated with direct prompting and CoT to test whether post-generation verification provides value beyond domain-specific pretraining at the same parameter scale. Frontier proprietary medical LLMs cannot be compared under the same controlled inference environment; this constraint is discussed in Section 5.3.

Direct inference (Direct): The base language model is prompted to generate answers directly without any augmentation, and serves as the unaugmented reference.

CoT (Wei et al., 2022): Few-shot examples guide the model to generate explicit step-by-step reasoning, serving as the standard baseline for reasoning-augmented approaches and remaining essential in related work (Liévin et al., 2024).

Self-Consistency (SC) (Wang et al., 2023): Multiple reasoning paths are sampled on top of CoT, and the final answer is determined by majority voting. This procedure tests whether agreement across sampled paths improves the final answer.

MindMap (Wen et al., 2024): A knowledge graph prompting framework published at ACL 2024. Knowledge graph triplet paths are converted into mind map representations and provided to the LLM, guiding the model to construct ontology-based graph reasoning.

Think-on-Graph (ToG) (Sun et al., 2024): An LLM-KG collaborative reasoning framework published at ICLR 2024. The LLM acts as an agent that iteratively performs beam search on the knowledge graph to discover reasoning paths, enhancing deep reasoning in a training-free, plug-and-play manner.

MedGraphRAG (Wu et al., 2024): A medical graph RAG framework proposed in 2024 that constructs a three-level knowledge graph and employs a U-shaped retrieval strategy to generate evidence-grounded responses.

Meditron-7B (Chen et al., 2023): A 7-billion-parameter LLM pretrained on a curated biomedical corpus comprising PubMed abstracts, clinical guidelines, and medical examination content. It is included to test whether KG-ACE's process-level verification provides additive value beyond domain-specific biomedical pretraining at the same parameter scale as the base model. Both Direct and CoT variants of Meditron-7B are evaluated.

4.4. Main results (RQ1)

Table 3 reports the results for RQ1. KG-ACE obtains the best value for every reported metric.

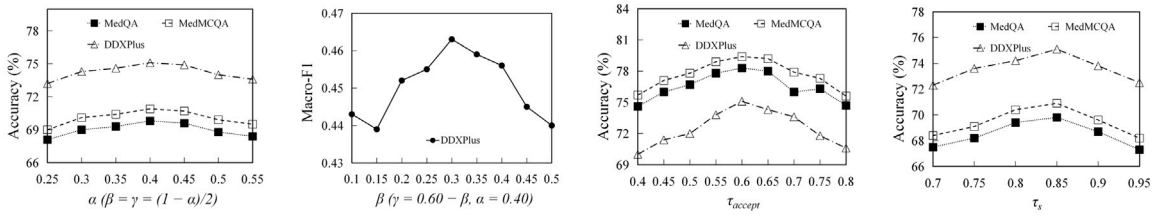
On MedQA and MedMCQA, Meditron-7B+CoT is the strongest baseline, with accuracies of 66.8% and 67.9%, respectively. KG-ACE improves both results by 3.0 percentage points. It also increases RCS by 0.050 on each dataset. The accuracy gains show that the verification pipeline adds value beyond biomedical pretraining, while the RCS gains show that independently sampled reasoning graphs are more reproducible. RCS does not measure correctness and is therefore interpreted only together with accuracy.

On DDXPlus, KG-ACE reaches 75.1% accuracy and a Macro-F1 of 0.463. Relative to ToG, these results correspond to gains of 4.8 percentage points in accuracy and 0.074 in Macro-F1. The relative Macro-F1 improvement (19.0%) is larger than the relative accuracy improvement (6.8%). This pattern suggests that explicit relation and logic checks are particularly useful when the output must cover several interacting diagnoses, not only identify one final label. Meditron-7B+CoT remains below ToG on this task, indicating that domain pretraining alone does not replace explicit constraint checking for multi-label diagnostic coverage.

On GenMedGPT-5k, BS-F1 varies within a narrow range (0.827–0.868), whereas LLM-Score varies from 0.628 to 0.791. The latter metric is therefore more discriminative in this experiment. KG-ACE achieves the best BS-F1 (0.868) and LLM-Score (0.791), exceeding ToG by 0.012 and 0.067, respectively. These results support the value of verifying relation validity and logical coherence in open-ended responses, although the limitations of the synthetic reference dialogues and LLM-based evaluation remain important (Section 5.3).

Table 4
Ablation study results. *w/o* denotes removal of the corresponding component. The last two rows are single-layer knowledge graph variants. All results are averages over three independent runs.

Variant	MedQA		GenMedGPT-5k		DDXPlus		
	Acc.	RCS	BS-F1	LLM-Score	Acc.	RCS	Macro-F1
KG-ACE (full)	**69.8**	**0.774**	**0.868**	**0.791**	**75.1**	**0.739**	**0.463**
w/o logic-level verif.	67.6	0.731	0.864	0.755	72.9	0.706	0.421
w/o relation-level verif.	66.4	0.718	0.861	0.739	71.4	0.692	0.398
w/o correction engine	65.8	0.711	0.859	0.731	70.6	0.683	0.386
w/o concept-level verif.	64.3	0.698	0.856	0.719	69.1	0.671	0.368
Consistency Knowledge Graph only	62.7	0.681	0.853	0.708	67.4	0.658	0.349
Ontology Knowledge Graph only	60.1	0.663	0.849	0.694	65.2	0.639	0.328



(a) Effect of concept weight α ($\beta = \gamma = (1-\alpha)/2$).

(b) Effect of relation weight β ($\gamma = 0.60 - \beta$).

(c) Effect of the acceptance threshold τ_{accept}.

(d) Effect of the concept-matching threshold τ_s.

Fig. 4. Sensitivity of KG-ACE to its four principal hyperparameters on MedQA, MedMCQA, and DDXPlus.

4.5. Ablation study (RQ2)

To address RQ2, each core component of KG-ACE is systematically removed in turn. Table 4 reports results on MedQA, GenMedGPT-5k, and DDXPlus, alongside two single-layer knowledge graph variants.

Concept-level verification has the largest individual effect. Removing it reduces MedQA accuracy by 5.5 percentage points, DDXPlus Macro-F1 by 0.095, and GenMedGPT-5k LLM-Score by 0.072. This result is consistent with the pipeline dependency: incorrect entity normalization propagates to both relation and logic verification. Removing the correction engine reduces MedQA accuracy by 4.0 percentage points, and removing relation-level verification reduces it by 3.4 percentage points. Logic-level verification has a smaller effect on MedQA accuracy (2.2 percentage points) but reduces DDXPlus Macro-F1 by 0.042, indicating a larger contribution to multi-label coverage.

The single-graph variants confirm that the two graph functions are complementary. Using only the Consistency Knowledge Graph reduces MedQA accuracy to 62.7%, while using only the Ontology Knowledge Graph reduces it to 60.1%. The corresponding DDXPlus Macro-F1 scores are 0.349 and 0.328. The Ontology Knowledge Graph is needed to establish which concepts a rule applies to, and the Consistency Knowledge Graph is needed to determine whether relations among those concepts are clinically compatible.

4.6. Hyperparameter sensitivity analysis (RQ3)

RQ3 examines the alignment weights (α, β, γ), the acceptance threshold τ_{accept}, and the concept-matching threshold τ_s. Fig. 4 reports validation results on MedQA, MedMCQA, and DDXPlus.

Performance is highest near $\alpha = 0.40$ and changes only modestly within $[0.35, 0.45]$. The validation grid search over $\alpha \in \{0.3, 0.4, 0.5\}$ and $\beta \in \{0.2, 0.3, 0.4\}$, with $\gamma = 1 - \alpha - \beta$, selects $(0.4, 0.3, 0.3)$. Reducing β to 0.10 lowers DDXPlus Macro-F1, whereas increasing it to 0.50 leaves too little weight for logic-level verification. The acceptance threshold shows an inverted-U pattern with its best value at 0.60: a lower threshold admits weakly aligned reasoning, while a higher threshold triggers unnecessary correction. Performance is comparatively stable for τ_s values up to 0.85 and then declines as valid medical mentions fail to match UMLS concepts. The same qualitative patterns appear on all three validation sets, supporting the use of one default configuration across tasks.

4.7. Case study (RQ4)

RQ4 is illustrated with a case in which a patient with hypertension is already taking aspirin. The original CoT reasoning includes the recommendation *Consider adding warfarin to reduce thrombotic risk*. Although the statement is linguistically plausible, it does not account for the high-risk interaction between warfarin and aspirin. KG-ACE matches the relation to a high-severity interaction edge in the Consistency Knowledge Graph ($\psi_C = 1.0$). The relation score falls from 0.82 to 0.51, which activates the correction engine.

The revised text reads: *Evaluate the indication for an appropriate single-agent anticoagulation strategy and monitor coagulation status*. This revision removes the unsupported co-administration while preserving the therapeutic objective.

The case demonstrates the intended mechanism: KG-ACE does not constrain the initial response, but it checks the completed reasoning against an explicit rule and retains a revision only when alignment improves. The example is illustrative rather than evidence of clinical safety or deployment readiness; those claims require prospective evaluation with clinicians.

5. Discussion

The results provide three main findings. First, the gains over Meditron-7B+CoT on MedQA and MedMCQA show that process-level verification contributes beyond domain-specific pretraining. Second, the larger improvement in DDXPlus Macro-F1 than in accuracy indicates that relation and logic checks are especially useful when several diagnoses must be considered jointly. Third, the ablation results show that concept normalization is a prerequisite for the later checks, while the two single-graph variants confirm that semantic grounding and clinical constraint checking are complementary.

5.1. Theoretical significance

KG-ACE separates two functions that are often combined implicitly: providing knowledge to a model and verifying how the model uses that knowledge. Retrieval determines which information is available before generation. Process-level verification evaluates whether a completed reasoning chain is compatible with an external specification after generation. The experimental comparison suggests that the second function can add value even when the base model has domain-specific pretraining or access to graph-based evidence.

The dual-layer architecture also separates semantic validity from constraint validity. The Ontology Knowledge Graph determines whether a mention refers to a recognized medical concept and whether its semantic type is appropriate. The Consistency Knowledge Graph determines whether relations among normalized concepts comply with clinical rules. An ontology-only representation cannot identify many unsafe interactions, whereas a constraint-only representation cannot reliably determine which concepts the rules apply to. The general design principle is therefore to represent domain meaning and operational constraints separately but evaluate them jointly at the point where an error occurs. This principle complements multi-granularity knowledge injection and explainable path-based graph reasoning (Gong et al., 2026; Li & Wu, 2025).

The analogy with formal verification is methodological, not literal. KG-ACE treats a reasoning step as a candidate state and a graph rule as a partial specification, but it does not provide the soundness or completeness guarantees of model checking over a closed transition system (Baier & Katoen, 2008). Medical reasoning is open-world, probabilistic, and dependent on graph coverage and extraction quality. A successful verification result therefore means that no violation was found within the available knowledge and extraction pipeline; it does not prove clinical correctness.

5.2. Practical significance and design recommendations

The measured latency and the absence of prospective clinical evaluation limit KG-ACE to clinician-supervised, non-time-critical decision support. Potential settings include differential-diagnosis review, medication reconciliation, and documentation assistance. The comparatively strong DDXPlus results favor cases in which several conditions or medications must be reconciled. They do not support autonomous use or bedside applications that require sub-second responses. A suitable workflow is to generate a draft, verify each detected violation, display the affected reasoning step together with the supporting graph edge and source, and require a clinician to accept or reject the proposed revision. This use complements medical-dialogue systems that improve access to preliminary guidance (Li et al., 2025); KG-ACE adds an audit layer when traceability and explicit constraint compliance are required.

The extraction analysis in Appendix B identifies entity linking as the largest measured error source. A deployment system should therefore disambiguate polysemous abbreviations before relation verification. For example, ‘‘MS’’ may denote multiple sclerosis or mitral stenosis, depending on the surrounding clinical context. Contextual embeddings, local mention windows, and explicit semantic-type checks can reduce such grounding errors. High-severity violations should be displayed with the relevant rule, confidence value, source, and affected reasoning step so that the clinician can inspect both the original statement and the proposed correction.

5.3. Threats to validity

Internal validity. The principal hyperparameters $(\alpha, \beta, \gamma, \tau_{\text{accept}}, \tau_s)$ were selected based on a grid search over the validation split of each dataset; it is possible that the reported sensitivity analysis underestimates sensitivity to parameter configurations not covered by the search grid. The ablation study removes one component at a time, which may not capture interaction effects between components.

Construct validity. The RCS metric measures reasoning stability, defined as path reproducibility under independent sampling, rather than reasoning correctness, and has not been validated against expert clinical judgment. It should be interpreted jointly with accuracy, BERTScore, and LLM-Score rather than as a standalone quality measure. BERTScore is known to have limited correlation with expert medical judgment and is treated as a reference baseline only, with LLM-Score used as the primary quality measure

for open-ended tasks. The alignment score weights ($\alpha = 0.4, \beta = 0.3, \gamma = 0.3$) are supported by dependency-based motivation and grid-search validation, but not by a formally derived optimality criterion, which remains an open theoretical question.

External validity. The most significant constraint on external validity is the absence of clinician-in-the-loop evaluation. The results reported in this paper establish the computational effectiveness of the verification and correction pipeline under automated evaluation conditions; they do not constitute evidence of clinical safety or deployment readiness, which would require prospective evaluation with clinical experts. This is identified as the highest-priority direction for future work. Additionally, two of the four evaluation datasets (GenMedGPT-5k and DDXPlus) are synthetic or semi-synthetic, and none contains real unstructured clinical notes or EHR-derived reasoning tasks. This is a meaningful limitation: real clinical text, characterized by ambiguous language, inconsistent terminology, and implicit inferential leaps, represents precisely the setting where KG-ACE's verification mechanisms would face the greatest challenge and potentially provide the greatest value. Whether the gains reported here transfer to such conditions remains an open empirical question. The framework is evaluated using a single base model (Qwen2.5-7B-Instruct); generalization to larger models and domain-specific base models requires further investigation. For frontier-scale models such as GPT-4-class systems or Med-PaLM 2, we expect the error distribution to shift rather than disappear. Stronger models may reduce concept-level normalization errors and obvious relation violations, while still benefiting from external checks for rare constraints, temporal ordering, and multi-condition interactions. The alignment weights may therefore require recalibration: a model with stronger entity grounding could warrant a lower concept-level weight and relatively higher relation- or logic-level weights. Correction prompts may also need to be more conservative to avoid unnecessary edits to already well-formed reasoning. These hypotheses require controlled evaluation with accessible frontier models and are left as future work. The effectiveness of KG-ACE also depends on the coverage quality of the knowledge graph. For rare diseases (fewer than 1 in 2000 prevalence) or treatments approved within the past 12 to 18 months, constraint edges may be absent, causing the framework to pass potentially unsafe reasoning unchallenged. Each verification decision is annotated with the source guideline and its publication date to support provenance tracking; integrating a retrieval step against regularly updated biomedical literature (e.g., PubMed) as a fallback is a planned future direction. Taken together, these dependencies mean that KG-ACE should therefore be interpreted as a bounded verification aid. A passed output indicates only that the available graph and extraction pipeline detected no violation; it does not establish that the reasoning is clinically correct.

Conclusion validity. BERTScore exhibits ceiling effects on the GenMedGPT-5k and MedQA tasks, spanning only 0.827 to 0.868 across all methods, which limits its sensitivity as a discriminative metric. Conclusions based on BERTScore differences should therefore be interpreted with caution, and the LLM-Score and accuracy metrics are treated as primary evidence. Relation extraction accuracy on the 200-sample MedQA subset achieves F1 = 78.9%; performance on longer, multi-hop reasoning chains was not separately measured and may differ from the MedQA subset. The DDXPlus findings should therefore not be interpreted as an estimate of performance under ideal extraction. Full error analysis is provided in Appendix B.

6. Conclusion

We introduced KG-ACE, a post-generation framework for verifying and correcting medical reasoning produced by LLMs. By separating concept standardization from clinical constraint checking through a dual-layer knowledge graph, KG-ACE evaluates each reasoning chain at the concept, relation, and logic levels and retains a correction only when the revised chain achieves a higher alignment score. This design makes the verification target explicit while preserving the original generation process. Experiments on four public benchmarks show that KG-ACE improves task performance, reasoning consistency, and knowledge-graph alignment relative to the evaluated baselines. The ablation results further indicate that concept normalization provides the foundation for reliable verification, whereas relation- and logic-level checks contribute complementary gains. In addition, the comparison with a domain-specific medical model suggests that process-level verification offers benefits beyond domain-specific pre-training. Nevertheless, these findings should be interpreted within the scope of the present evaluation. The study does not include prospective clinician assessment, depends on the coverage and quality of the underlying knowledge graph, and evaluates only a single base LLM. Future work should therefore investigate clinician-in-the-loop use on real or de-identified clinical cases, extend the knowledge graph with regularly updated biomedical evidence, and examine whether the verification weights and correction policy transfer across different model families and clinical tasks. KG-ACE is intended as an auditable decision-support layer rather than an autonomous clinical decision maker; a reasoning chain that passes verification indicates only that no violation has been detected within the available knowledge graph and extraction pipeline, rather than that the resulting conclusion is clinically correct.

CRediT authorship contribution statement

Chen Sun: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Junhua Fang:** Writing – review & editing, Supervision, Investigation, Funding acquisition, Conceptualization. **Li Ni:** Resources, Data curation. **Kaiwen Yu:** Resources, Data curation. **Yating Lu:** Resources, Data curation. **Yan Lu:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Computational overhead analysis

Table A.1 reports wall-clock inference latency, average correction iterations, and GPU memory use on MedQA. Measurements were obtained on one NVIDIA A800-80GB GPU with batch size 1 and are averaged over 200 randomly sampled test queries and three repetitions.

KG-ACE incurs a mean latency of 4.52 s per query, averaged over 2.3 correction iterations. This is 34% higher than ToG (3.36 s/q) and 23% higher than MedGraphRAG (3.68 s/q). Compared with Meditron-7B+CoT (0.64 s/q), KG-ACE requires substantially more time per query, a cost attributable to the iterative verification-correction loop rather than to the base model itself; the Direct and CoT variants of both Qwen2.5-7B and Meditron-7B exhibit comparable per-query latency (0.38–0.64 s/q), confirming that the overhead stems from the KG-ACE pipeline rather than model-specific inference cost. The additional latency relative to retrieval-based baselines derives from two sources: (a) the three-level verification pipeline, which performs named entity recognition, triplet extraction, and graph matching sequentially; and (b) the correction loop, which re-runs the alignment computation after each corrective prompt. The per-iteration verification overhead excluding LLM generation averages 0.57 s, indicating that the dominant latency cost remains LLM generation itself rather than KG query operations.

The measured latency may be compatible with asynchronous review, but workflow acceptability must be established in the intended clinical setting rather than inferred from a universal time threshold. For lower-latency settings, possible optimizations include caching frequently used graph substructures, using approximate nearest-neighbor matching for candidate retrieval, and terminating correction when the marginal alignment gain falls below a secondary threshold ϵ'.

Appendix B. Relation extraction error analysis

To characterize the accuracy of the relation extraction module and assess how extraction errors propagate through the verification pipeline, we randomly sampled 200 reasoning texts from the MedQA test set and manually annotated the ground-truth triplet sets. Three annotators independently labeled each sample, and disagreements were resolved by majority vote; inter-annotator agreement reached Fleiss' $\kappa = 0.81$, indicating substantial agreement.

The dominant error category is entity linking errors (41%), where one entity in a triplet is incorrectly matched to a UMLS concept, typically due to polysemy. For example, the abbreviation "MS" may be resolved to Multiple Sclerosis rather than Mitral Stenosis in cardiological contexts, introducing an incorrect entity into the verification pipeline. Boundary errors (23%) arise when an entity or relation span is only partially identified or includes unrelated tokens. Relation-type confusions, such as labeling a causal relation as temporal, are counted with the affected extraction error rather than described as span-boundary errors. Missing relations (36%) occur when implicit clinical inferences are not explicitly expressed as subject–predicate–object structures in the reasoning text, leaving valid constraints undetected by the extraction module.

Table A.1
Computational overhead comparison on MedQA (NVIDIA A800-80GB, batch size = 1). Latency is reported as mean ± standard deviation over 200 queries. Iters: average number of correction iterations (N/A: not applicable to non-iterative methods).

Method	Avg. Iters	Latency (s/q)	GPU Mem (GB)	Acc. (%)
Direct	N/A	0.38 ± 0.05	14.6	53.4
CoT	N/A	0.61 ± 0.07	14.6	58.9
Self-Consistency	5	2.43 ± 0.22	14.6	61.7
MindMap	N/A	2.17 ± 0.26	15.4	63.4
ToG	3.2	3.36 ± 0.38	16.2	65.1
MedGraphRAG	2.8	3.68 ± 0.41	16.8	64.2
Meditron-7B (Direct)	N/A	0.42 ± 0.05	14.6	59.3
Meditron-7B + CoT	N/A	0.64 ± 0.08	14.6	66.8
KG-ACE	**2.3**	4.52 ± 0.53	**17.8**	**69.8**

Table B.1
Relation extraction performance on 200 manually annotated MedQA reasoning texts, and downstream impact on KG-ACE accuracy when gold triplets replace automatic extraction. N/A indicates that the column is not applicable to the corresponding row.

Metric	Value	Error category (%)
Precision	81.3%	N/A
Recall	76.8%	N/A
F1	78.9%	N/A
Boundary errors	N/A	23%
Entity linking errors	N/A	41%
Missing relations	N/A	36%
KG-ACE (auto extraction)	70.4% Acc.	N/A
KG-ACE (gold triplets)	73.1% Acc.	N/A
Δ (gold − auto)	+2.7 pp	N/A

The gold-triplet ablation reported in Table B.1 shows that replacing automatic extraction with manually annotated triplets raises KG-ACE's accuracy by 2.7 percentage points on the 200-sample subset, from 70.4% to 73.1%. This gap indicates that relation extraction errors account for a moderate but not dominant fraction of the total error budget, and that the verification pipeline maintains partial robustness to extraction noise owing to redundant multi-path coverage in the reasoning graph. Extraction performance on longer, multi-hop DDXPlus reasoning chains was not measured separately and may differ from the 78.9% F1 observed on the MedQA subset. Improving medical relation extraction, particularly for polysemous entities and implicit clinical inferences, is therefore identified as a key direction for future work.

Appendix C. Prompt templates for the correction engine

This appendix provides the verbatim prompt templates used in the three correction modes of the intelligent correction engine (Section 3.3). All templates follow a structured format: a system-role declaration, a runtime-populated context block, and a constrained instruction block. Variables enclosed in square brackets are substituted at runtime; substitution procedures are described at the end of this appendix.

Local Correction Prompt (LocalPrompt)

System: You are a clinical language normalization assistant. Your task is to correct non-standard medical terminology in a reasoning chain while preserving its overall structure and meaning.

Context:

- *Reasoning text:* `[R*]`
- *Non-standard term detected:* `[X]`
- *UMLS-normalized equivalent:* `[Y]`

Instruction: The clinical term *[X]* in the reasoning path has been identified as non-standard. Replace it with the UMLS-normalized equivalent *[Y]* and adjust adjacent clauses for grammatical consistency. Modify only the affected phrase and leave all other content unchanged. Return the corrected reasoning text only, without explanation.

Structural Correction Prompt (StructuralPrompt)

System: You are a clinical reasoning revision assistant. Your task is to resolve a specific constraint violation in a reasoning chain by revising the affected step and its logically dependent downstream steps.

Context:

- *Reasoning text:* `[R*]`
- *Violated step index:* Step `[k]`
- *Asserted relation:* (`[e1]`, `[r]`, `[e2]`)
- *Constraint value:* ψ_C = `[value]`
- *Violation type:* `constraint_violation`

Instruction: Step *[k]* of the reasoning chain asserts the relation (e_1, r, e_2), which has been flagged as clinically prohibited (ψ_C = *[value]*). Revise step *[k]* and its downstream dependencies to eliminate this constraint violation while preserving the overall clinical context and logical flow. Do not modify steps that are not logically dependent on step *[k]*. Return the corrected reasoning text only, without explanation.

Global Correction Prompt (GlobalPrompt)

System: You are a clinical reasoning regeneration assistant. Your task is to regenerate a complete reasoning chain that satisfies a given set of clinical constraints and is internally logically consistent.

Context:

- *Original reasoning text:* `[R*]`
- *Detected violations:* `[list of violations]`, each specifying violation type, involved entities, and constraint description
- *Relevant KG subgraph:* entities $\mathcal{V}(\mathcal{G}_R)$ and their constraint edges from $\mathcal{G}_C$

Instruction: The following logical inconsistencies were detected in the reasoning chain: *[list of violations]*. Regenerate the complete reasoning chain such that: (1) each of the listed constraints is satisfied, (2) all clinical logic is internally consistent, and (3) the conclusion is preserved where clinically defensible. Do not introduce new clinical claims beyond those supported by the provided KG subgraph. Return the regenerated reasoning text only, without explanation.

Table C.1
Runtime variables used in the correction prompt templates. Sources refer to the component or algorithm that produces each value at inference time.

Variable	Description	Source
`[R*]`	Current reasoning text at iteration t	Algorithm 4, line 3. It is initialized as R and updated to R_{temp} when the increase in the alignment score exceeds ϵ at line 15.
`[X]`	Non-standard clinical term identified in R^*	Concept-level verification using named entity recognition and $\text{match}(m, c)$ against the node set V_O of the Ontology Knowledge Graph (Section 3.2).
`[Y]`	UMLS-normalized equivalent of `[X]`	Ontology Knowledge Graph $\mathcal{G}_O$. The normalized term is retrieved using $\text{Standardize}(m)$ (Section 3.1).
`[k]`	Index of the reasoning step containing the detected violation	The `step_id` field of the `constraint_violation` record produced by relation-level verification. The value is passed to StructuralPrompt in Algorithm 3, line 8.
`[e1]`, `[r]`, `[e2]`	Entities and relation type in the flagged triple (e_1, r, e_2)	Relation extraction module. The extracted triple is matched against the relation sets $E_O \cup E_C$ (Section 3.2).
`[value]`	Constraint confidence ψ_C associated with the violated edge	Consistency Knowledge Graph $\mathcal{G}_C$, in which $\psi_C \in [0, 1]$ is assigned during graph construction (Section 3.1).
`[list of violations]`	Structured list of detected violations, including the violation type, involved entities, affected reasoning steps, constraint description, and severity rating	Output of the violation detection function $\mathcal{V}(\mathcal{G}_R)$ during logic-level verification (Algorithm 2). The list covers `temporal_conflict`, `causal_gap`, and `contradiction` records.
$\mathcal{G}_{\text{sub}}$	Relevant KG subgraph containing the entities and constraint edges associated with the detected violations	Extracted from the Consistency Knowledge Graph $\mathcal{G}_C$ during global correction. The subgraph limits the scope of regenerated clinical claims (Algorithm 3, line 1; Section 3.3).

Runtime variable substitution

Table C.1 summarizes the runtime variables used in the three correction prompt templates. The variables are populated automatically by the corresponding KG-ACE components before each LLM invocation.

All runtime variables are substituted programmatically before the corresponding prompt is submitted to the LLM. No manual editing is performed during inference. For local correction, the system inserts the current reasoning text, the detected non-standard term, and its normalized UMLS equivalent. For structural correction, it additionally inserts the affected step index, the flagged relation triple, and the associated constraint confidence. For global correction, the system provides the complete list of detected violations together with the relevant subgraph extracted from $\mathcal{G}_C$.

The correction engine therefore receives only the evidence required for the selected correction mode. This design restricts unnecessary modifications, preserves unaffected reasoning steps whenever possible, and prevents the global correction prompt from introducing clinical claims that are not supported by the supplied knowledge graph.

Data availability

Data will be made available on request.

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